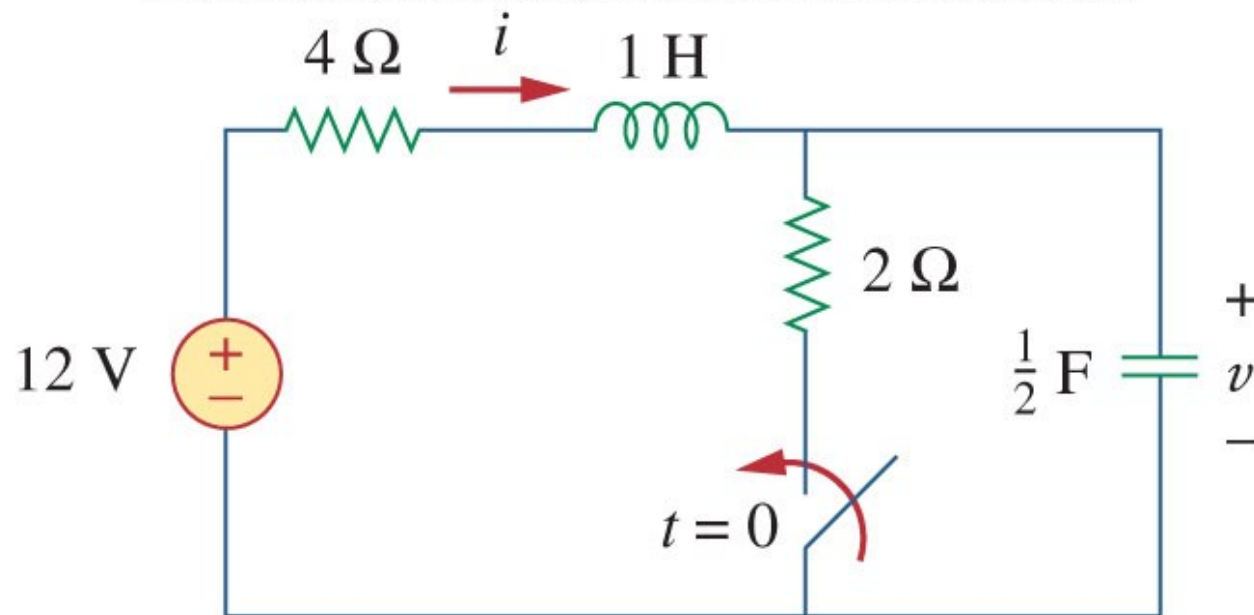




General Second-Order Circuits

- An example

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General Second-Order Circuits

- The principles of solving the series/parallel forms of RLC circuits can **be applied/extended** to **general second-order circuits**, by taking the following six steps:
 - First determine the initial conditions, $x(0)$ and $dx(0)/dt$.
 - Applying KVL and KCL**, to find a **general second-order differential equation** about $x(t)$.
Then solve the equation:
 - Depending on the roots of C.E. , the form of general solution $x_{g.s.}(t)$ (3 cases) of homogeneous equation can be determined.**
 - We obtain the **particular solution** by observation/calculation, **specially** for a DC/step response $x_{p.s.}(t)=x(\infty)$
 - The complete response = general solution + particular solution.
$$x(t)=x_{P.S.}(t)+ x_{G.S.}(t)$$
 - Using the initial conditions to determine the constants of $x(t)$.



$x(t)$ = unknown variable (voltage or current)

Differential equation:

$$x'' + ax' + bx = c$$

Initial conditions:

$$x(0) \text{ and } x'(0)$$

Final condition:

$$x(\infty) = \frac{c}{b}$$

$$\alpha = \frac{a}{2}$$

$$\omega_0 = \sqrt{b}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Overdamped Response $\alpha > \omega_0$

$$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)]$$

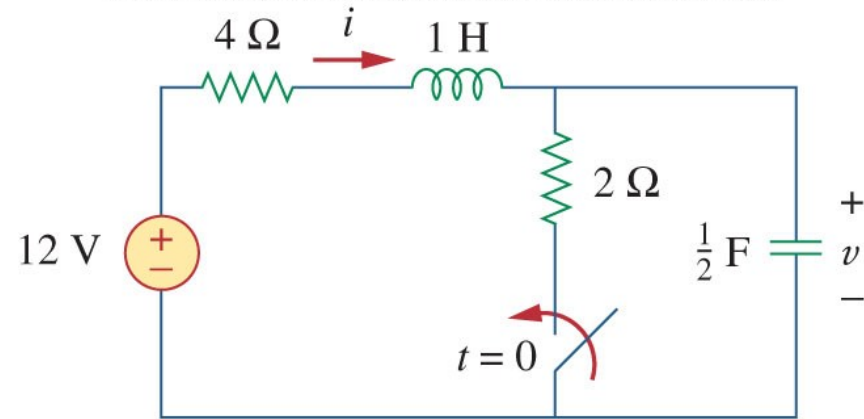


General RLC Circuits

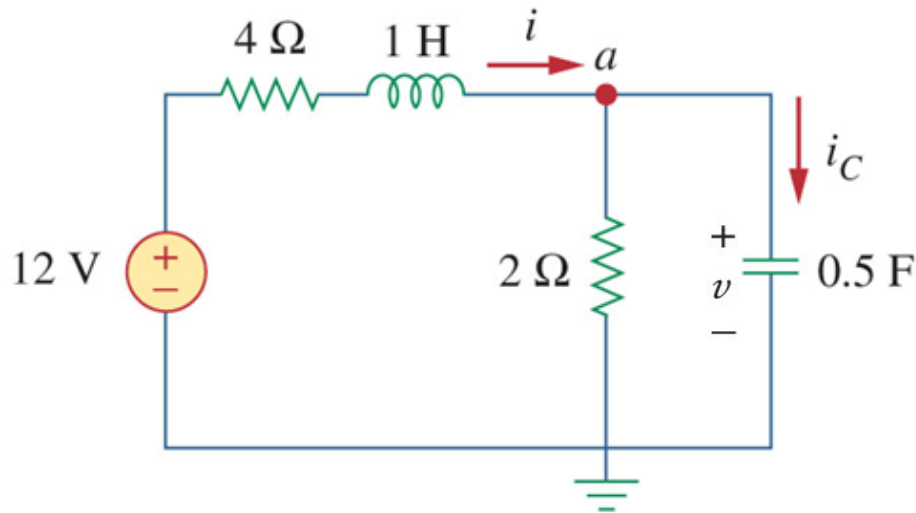
- Find the complete response $v(t)$ for $t > 0$ in the circuit.

1. Initial conditions

$$v(0^+) = v(0^-) = 12V, i(0^+) = i(0^-) = 0$$



$t = 0$

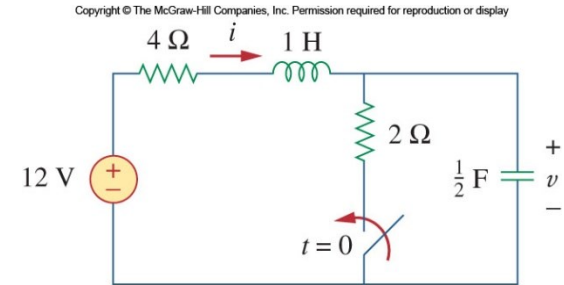
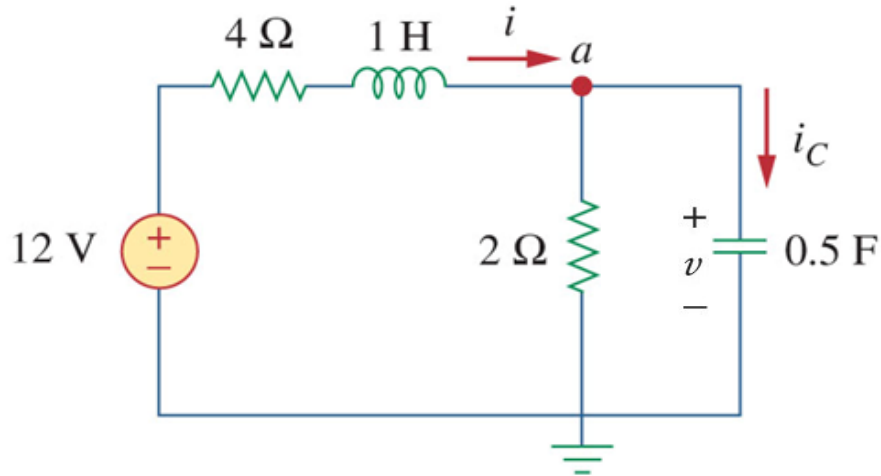


$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = \frac{-12/2}{0.5} = -12 V/s$$



General RLC Circuits

- Find the complete response $v(t)$ for $t > 0$



- To find a **general** second-order differential equation about $x(t)$



$$\frac{d^2v}{dt^2} + 5 \frac{dv}{dt} + 6v = 24$$

3. General Solution:

➡ General Solution $v_t(t) = A_1 e^{-2t} + A_2 e^{-3t}$

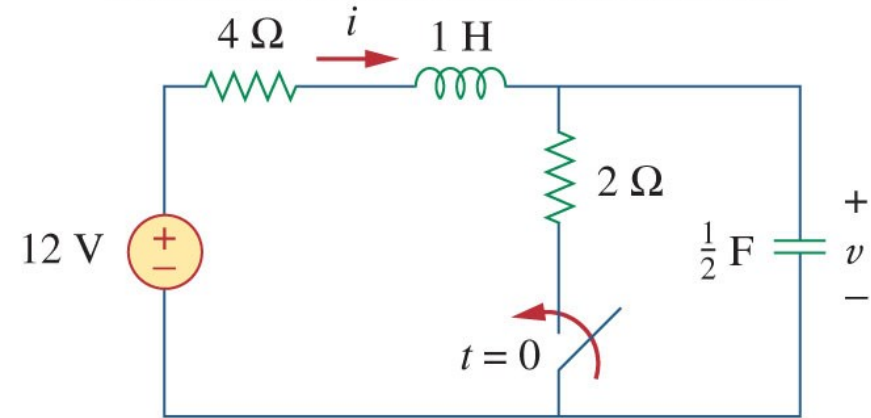
4. Particular Solution : Steady-state response $v_{ss}(t) = 4V$

5. Put together : $v(t) = 4 + A_1 e^{-2t} - A_2 e^{-3t}$

6. Using initial conditions to determine A_1 , A_2



- Use mesh analysis method to find the complete response $i(t)$ for $t > 0$ in the circuit.

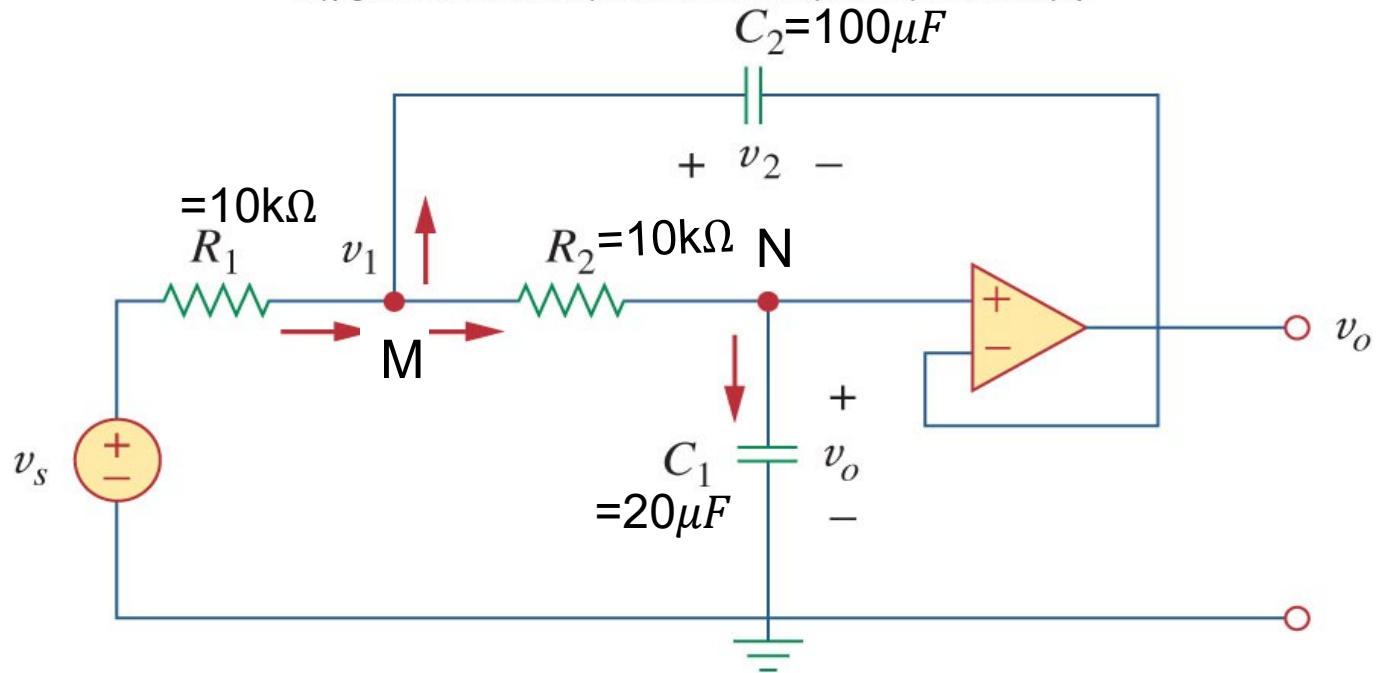




Example of 2nd-order op-amp circuits

- Find v_o for $t > 0$
when $v_s = 10u(t)mV$.

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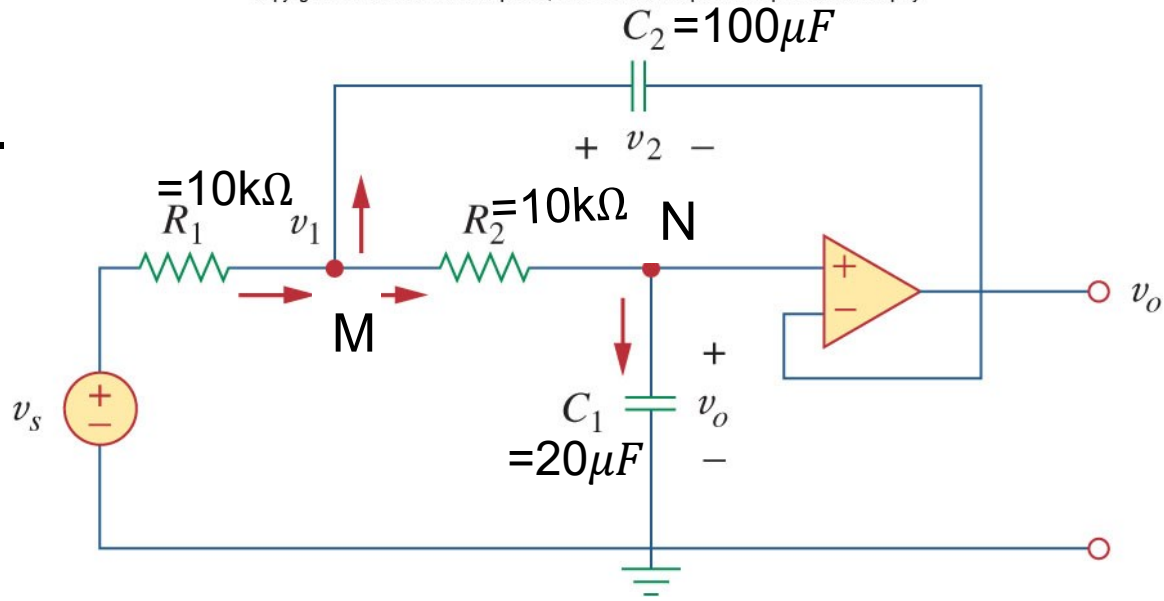
$$\text{Initial conditions: } v_o(0^+) = 0, C_1 \frac{dv_o(0^+)}{dt} = \frac{v_1(0^+) - v_o(0^+)}{R_2} = \frac{v_2(0^+)}{R_2} = 0$$



Example of 2nd-order op-amp circuits

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- Find v_o for $t > 0$
when $v_s = 10u(t)mV$.



KCL at node M:

$$\frac{v_s - v_1}{R_1} = C_2 \frac{dv_2}{dt} + \frac{v_1 - v_o}{R_2}$$

KCL at node N:

$$C_1 \frac{dv_o}{dt} = \frac{v_1 - v_o}{R_2}$$

and we have $v_1 - v_2 = v_o$

$$\Rightarrow \frac{d^2 v_o}{dt^2} + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} \right) \frac{dv_o}{dt} + \frac{v_o}{R_1 R_2 C_1 C_2} = \frac{v_s}{R_1 R_2 C_1 C_2}$$

